POCKETSMART AI

Your Smart Budget & Recommendation Assistant

AI + BUDGET + PERSONALIZATION

One platform • Three smart planners • Better decisions

Home Interior | Party Planning | Jewelry Matching

Project Presentation

The Problem

Budget planning becomes difficult when choices are scattered across many platforms.

TOO MANY CHOICES

Different products, vendors, styles and prices make co

BUDGET UNCERTAINTY

Users often know their total budget—but not how to di

NO PERSONALIZATION

Generic search results rarely consider occasion, style, quar

PocketSmart AI turns these problems into one guided, intelligent workflow.

Our Solution

A single assistant that understands the user's need before recommending options.

INPUT

Budget
Preferences
Quantity
Occasion
Optional image

AI BRAIN

Gemini
context analysis
+ prompt logic
+ budget awareness

RECOMMEND

Relevant products
Category-wise plan
Platform-aware options
Budget-friendly choices

Goal: Make planning faster, clearer and smarter.

Three Smart Planners

Each planner uses the same AI engine with domain-specific inputs.

🏠 HOME INTERIOR

Budget
Rooms & quantities
Style preferences

Output: furniture, lighting, decor ar

🎉 PARTY PLANNER

Total budget
Guests & event type
Venue details

Output: food, venue, decoration an

💎 J EWELRY MATCH

Budget
Occasion & style
Optional outfit image

Output: jewelry suggestions matched

Scenario 1 —Home Interior

Example: plan a room without crossing the total budget.

STEP 1

Budget: ₹50,000
Rooms: Living + Bedroom
Items: lights, fan, table
Style: Modern

STEP 2

AI understands:

- item quantities
- priorities
- style
- price limits

STEP 3 —SMART OUTPUT

Category-wise allocation

Lighting → budget range
Furniture → budget range
Decor → budget range

Then show suitable options from connected/ avail

Scenario 2 —AI Party Budget

The assistant converts one total budget into a practical event plan.

01

EVENT

Birthday / Wedding / Corporate

02

BUDGET

Total amount + guest count

03

ALLOCATE

Food + venue + decor + entertainment

04

RECOMMEND

Options matched to the plan

Result: less guesswork, better budget control.

Scenario 3 — Jewelry Recommendations

Optional outfit image input adds visual context to the recommendation.

USER INPUT

Budget
Occasion
Preferred style
Outfit image (optional)

AI ANALYSIS

Understands text context
+ analyzes visual cues
+ considers color/style coordinatio

OUTPUT

Matching jewelry ideas
Budget-friendly choices
Occasion-appropriate styles
Platform/product references where a

System Architecture

A modular design keeps the frontend, API layer and AI logic easy to maintain.

FRONTEND	HTML + CSS + JavaScript + Jinja2
FASTAPI BACKEND	Routes • Auth • Sessions • Validation
AI SERVICE	Gemini integration • Prompt orchestration • Image handling
DATA / SOURCES	Product & service APIs / mock integrations

Technology Stack

Core technologies used to build the prototype.

FastAPI

API & backend routing

Gemini

AI reasoning & multimodal input

Python

Application logic

HTML / CSS / JS

Interactive frontend

Jinja2

Server-rendered templates

JWT + Sessions

Authentication & personalization

Backend API Map

Clear endpoints connect each user action to the correct service.

`/generate-home`

Home recommendations

`/token`

JWT access token

`/generate-party`

Party recommendations

`/session-info / session-data`

Session & personalization

`/generate-jewelry`

Jewelry + optional image

`/history`

Past recommendation queries

`/register / login / logout`

Authentication

`/recommendations-details`

Detailed recommendation view

How One Recommendation Is Generated

From user input to a useful, budget-aware result.

1

Collect

Budget + preferences

2

Validate

Check required fields

3

Build prompt

Create domain-specific co

4

Ask AI

Generate structured sugg

5

Filter

Check budget & relevance

6

Display

Show clear recommendatio

Development Milestones

A simple roadmap from AI setup to a tested application.

01	Gemini Initialization	API access • text test • image test
02	Core Functionality	Planner logic • prompts • authentication
03	FastAPI Integration	Routes • CORS • sessions • startup
04	UI Development	Forms • dashboards • recommendation cards
05	Testing & Optimization	Real-world cases • validation • fallback

User Experience —From Login to Recommendation

The UI is organized around a short, intuitive journey.

01 ster / Login

Secure entry

02 nboard

Recent plans & shortcuts

03 ose Planner

Home • Party • J ewelry

04 r Details

Budget + preferences

05 v Results

AI recommendation cards

06 ory

Review & reuse plans

Testing & Optimization

Quality is measured by usefulness—not just whether the API responds.

FUNCTIONAL TESTS

✓ Valid budgets
✓ Missing fields
✓ Different planner types
✓ Login / logout
✓ Image + text input

AI QUALITY CHECKS

✓ Budget adherence
✓ Relevance
✓ Style / occasion fit
✓ Structured output
✓ Fallback when results are weak

UX CHECKS

✓ Mobile-friendly forms
✓ Clear recommendation cards
✓ Fast navigation
✓ Session continuity
✓ History & reuse

Why PocketSmart AI?

The project brings planning, personalization and budget awareness into one experience.

SMART

AI-assisted decisions

BUDGET-AWARE

Recommendations stay aligned

PERSONALIZED

Context, style and occasion ma

MODULAR

Easy to extend to new planners

SCALABLE

Can connect to more data source

From “What should I buy?” to “What fits my budget and need best?”

Conclusion

PocketSmart AI is a practical example of generative AI solving everyday planning problems.

- Combines budget inputs, preferences and context into one recommendation flow.
- Uses separate planners for Home Interior, Party Planning and Jewelry Matching.
- FastAPI provides a clean backend for routing, authentication, sessions and AI integration.
- Future scope: live product APIs, price tracking, saved budgets, notifications and more planner categories.

PLAN SMART • SPEND SMART • CHOOSE SMART

Thank You